

Tax, Tip, Discount and Markup

Tax and tip added on, discounts and sale prices, and markups worked back to the original price

Directions:

- Every amount of money on this page is in **dollars**, and every percent is a percent *of the price named in the same sentence* — read which price that is before you multiply.
- Adding on (tax, tip, markup) multiplies by $1 + \frac{p}{100}$. Taking off (discount) multiplies by $1 - \frac{p}{100}$.
- Round money answers to the nearest cent and write the cents digits even when they are zeros.

1. A book is priced at 24 dollars and the sales tax rate is 5%.

(a) the sales tax, in dollars: _____

(b) the total paid, in dollars: _____

2. A meal costs 40 dollars before the tip, and the diners leave an 18% tip on that amount.

(a) the tip, in dollars: _____

(b) the total paid, in dollars: _____

3. A jacket is marked 60 dollars and the rack says 25% off.

(a) the discount, in dollars: _____

(b) the sale price, in dollars: _____

4. A bicycle repair is billed at 132 dollars before tax, and the tax rate is 6.5%. What is the total bill, in dollars?

Answer: _____

5. A shop pays 35 dollars for a lamp and marks it up 40% of what it paid.

(a) the markup, in dollars: _____

(b) the selling price, in dollars: _____

6. A board game was priced at 48 dollars and is now selling for 36 dollars. What percent was taken off the original price?

Answer: _____

7. A group leaves a 15% tip on a restaurant bill of 86 dollars. How much is the tip, in dollars?

Answer: _____

8. A chair sells for 84 dollars, which is what the shop paid for it plus a 20% markup on that amount. How many dollars did the shop pay for the chair?

Answer: _____

9. A coat is on sale for 68 dollars after a 15% discount. What was the price, in dollars, before the discount?

Answer: _____

10. A tent is marked 120 dollars. It is on sale at 30% off, and a sales tax of 8% is charged on the sale price.

(a) the sale price, in dollars: _____

(b) the sales tax, in dollars: _____

(c) the total paid, in dollars: _____

(d) Would charging the tax first and taking the discount second change the total? Explain in the work space.